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Noise as Information for Illiquidity

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1 Big picture

- **Question.** Can the cross-sectional “noise” in U.S. Treasury prices be used as a market-wide measure of illiquidity, and is this liquidity risk priced in assets that depend on arbitrage capital?
- **Setting.** The paper studies daily U.S. Treasury prices from 1987–2011, hedge fund returns from TASS from 1994–2011, and currency carry trade returns from 1987–2011.
- **Core idea.** When arbitrage capital is abundant, relative-value traders smooth the Treasury yield curve and eliminate small pricing deviations. When arbitrage capital is scarce, yields in different maturity “habitats” move more freely, and the fitted yield curve becomes noisy.
- **Main measurement result.** The paper constructs a daily **noise measure** as the root mean squared deviation between observed Treasury yields and yields implied by a fitted zero-coupon curve.
- **Why Treasuries?** The Treasury market is central, highly liquid, and close to default-free. If even Treasury prices show abnormal cross-sectional mispricing, that is a strong signal of a broad shortage of arbitrage capital rather than just local market frictions.
- **Main time-series result.** The noise measure spikes in major liquidity episodes: the 1987 crash, LTCM, September 11, Bear Stearns, Lehman, and the broader 2008–2009 crisis.
- **Main asset-pricing result.** Hedge funds with more negative noise betas earn higher average returns. In Fama–MacBeth tests, the estimated noise risk premium is negative and significant, meaning that exposure to bad liquidity states earns compensation.
- **Carry trade result.** High-interest-rate “asset” currencies have more negative noise betas than low-interest-rate funding currencies. Adjusting for noise risk greatly reduces the apparent excess return of carry trades.
- **Economic takeaway.** “Noise” is not just measurement error. In this paper, Treasury yield-curve noise is informative about the aggregate supply of arbitrage capital and therefore about market-wide liquidity risk.

2 Data and measurement (key objects)

2.1 Treasury data

- **Primary source.** The Treasury data come from the CRSP Daily Treasury database.
- **Sample period.** The paper uses daily Treasury prices from 1987–2011.
- **Securities included.** The sample contains Treasury bills, notes, and bonds that are:
 - noncallable;

- nonflowering;
- not subject to special tax treatment.
- **Cleaning rules.** The authors remove observations with obvious pricing errors, including negative prices, negative yields, or negative bid-ask spreads.
- **Maturity filters.**
 - Securities with remaining maturity less than 1 month are dropped because the short end can be affected by special liquidity problems.
 - Securities longer than 10 years are dropped because fewer observations make the fitted curve less reliable.
 - The yield curve is fitted using maturities from 1 month to 10 years.
 - The noise measure itself is constructed using bonds with maturities from 1 year to 10 years.
- **Typical cross-section.** On average, there are about **163** Treasury securities per day for curve fitting and about **109** bonds with maturity 1–10 years for constructing the noise measure.
- **Typical bond characteristics.** In the full sample, the bonds used for noise construction have average maturity of about **3.85 years**, average age of **3.96 years**, average duration of **3.29 years**, average bid-ask yield spread of **2.11 basis points**, and average yield of **4.90 percent**.

2.2 Noise measure

- The paper first backs out a smooth zero-coupon yield curve each day.
- It then prices each Treasury bond using the fitted curve.
- For each bond, the key object is the difference between its observed market yield and its model-implied yield.
- These deviations are aggregated across bonds using root mean squared error.
- The resulting daily variable is called **Noise**.
- The paper uses the word “noise” in the fixed-income sense: deviations from a pricing model are often called noise or pricing errors.
- Important interpretation: the paper does *not* treat this noise as useless error. Instead, it argues that large Treasury pricing errors are a symptom of scarce arbitrage capital.

2.3 Other liquidity and market variables

The paper compares the noise measure with several existing market variables:

- **Treasury variables:** 3-month Treasury bill rate, term spread, bond return volatility.
- **Treasury liquidity measures:** 5-year and 10-year on-the-run premiums, and the RefCorp-Treasury spread.
- **Stock market variables:** CRSP value-weighted stock return, VIX, and the Pastor–Stambaugh equity liquidity innovation.
- **Funding and credit variables:** overnight general collateral repo rate, LIBOR-Treasury bill spread, and Baa-Aaa default spread.

2.4 Hedge fund data

- **Source.** Hedge fund data come from the Lipper TASS database.
- **Sample period.** The hedge fund tests use 1994–2011 because the TASS graveyard database begins in 1994, which helps reduce survivorship bias.
- **Filters.** The sample includes funds that:
 - report net-of-fee monthly returns in U.S. dollars;
 - have assets under management of at least \$10 million;
 - have at least 24 months of return history.
- **Final hedge fund sample.** The sample contains **5,392** hedge funds, including **3,557** graveyard funds.
- **Average characteristics.** Across the full sample, average monthly return is about **0.67 percent**, average monthly return volatility is about **3.68 percent**, average AUM is about **\$160 million**, and average monthly return autocorrelation is about **0.19**.

2.5 Currency carry trade data

- The paper follows Lustig, Roussanov, and Verdelhan and studies **36 currencies** from developed and emerging countries.
- Spot and 1-month forward exchange rates come from Barclays and Reuters via Datastream.
- The sample period is 1987–2011.
- Each month, currencies are sorted into six portfolios by forward discount, which is closely related to the interest-rate differential under covered interest parity.
- Portfolio 1 contains high-interest-rate target currencies; portfolio 6 contains low-interest-rate funding currencies.

2.6 Unobservables and interpretation

- **Unobserved object of interest.** The true object is aggregate arbitrage capital or market-wide liquidity supply, which is not directly observable.
- **Measurement proxy.** Treasury yield-curve noise proxies for this object because scarce arbitrage capital allows cross-sectional pricing errors to persist.
- **Demand-versus-supply issue.** A single bond's pricing error could reflect either liquidity demand or liquidity supply. The paper argues that averaging across many Treasury securities mostly washes out idiosyncratic liquidity demand, while a broad shortage of arbitrage capital affects the entire curve.
- **Timing.** Daily noise is used to describe crisis episodes; monthly changes in noise are used for regressions and asset-pricing tests.

3 Models and empirical specifications (all equations + interpretation)

3.1 Limits-of-arbitrage intuition

The conceptual mechanism is simple:

- Treasury bonds have prices disciplined by the yield curve.
- Relative-value traders arbitrage deviations between individual bond prices and the fitted curve.
- When arbitrage capital is plentiful, deviations are quickly traded away.
- When capital is scarce, deviations persist and the yield curve becomes noisier.

Interpretation. The noise measure is an inverse proxy for the strength of arbitrage forces. High noise means the market is less able or less willing to deploy capital to eliminate relative-value opportunities.

3.2 Svensson yield-curve model

The baseline curve-fitting model is the Svensson model for the instantaneous forward rate:

$$f(m, b) = \beta_0 + \beta_1 \exp\left(-\frac{m}{\tau_1}\right) + \beta_2 \frac{m}{\tau_1} \exp\left(-\frac{m}{\tau_1}\right) + \beta_3 \frac{m}{\tau_2} \exp\left(-\frac{m}{\tau_2}\right), \quad (1)$$

where:

- m is time to maturity;

- $b = (\beta_0, \beta_1, \beta_2, \beta_3, \tau_1, \tau_2)$ is the parameter vector;
- β_0 captures the long-horizon forward rate;
- $\beta_0 + \beta_1$ captures the short-end forward rate;
- (β_2, τ_1) and (β_3, τ_2) govern the humps in the forward curve.

Interpretation. The yield curve is summarized by a small number of factors. This makes Treasury prices a useful setting because pricing deviations can be measured relative to a transparent benchmark.

3.3 Daily curve fitting

On each day t , the model parameters are estimated by minimizing duration-weighted pricing errors:

$$b_t = \arg \min_b \sum_{i=1}^{N_t} \left[\left(P^i(b) - P_t^i \right) \frac{1}{D_i} \right]^2, \quad (2)$$

where:

- N_t is the number of Treasury securities available for curve fitting;
- P_t^i is the market price of security i ;
- $P^i(b)$ is the model-implied price under parameter vector b ;
- D_i is Macaulay duration.

Interpretation. Weighting price errors by inverse duration makes the objective close to minimizing yield errors. This is useful because the noise measure is ultimately expressed in yield basis points.

3.4 Noise measure

Given the fitted curve parameters b_t , the daily noise measure is

$$Noise_t = \sqrt{\frac{1}{N_t} \sum_{i=1}^{N_t} (y_t^i - y^i(b_t))^2}, \quad (3)$$

where:

- y_t^i is the observed yield of bond i on day t ;
- $y^i(b_t)$ is the model-implied yield from the fitted curve;
- the summation is over bonds with remaining maturity between 1 and 10 years.

Interpretation. This is the cross-sectional dispersion of Treasury yields around the fitted curve. A higher value means that individual bond yields are more misaligned relative to a smooth term structure.

3.5 Outlier filter

The authors also apply a mild cross-sectional filter:

- compute cross-sectional dispersion in yield pricing errors;
- exclude bonds whose yield-to-maturity is more than four standard deviations from the model yield;
- then compute the noise measure from the remaining bonds.

Interpretation. The filter prevents one or two bad quotes from driving the index. In practice it removes very few observations: over 1987–2011, there is no day on which it removes more than four bonds.

3.6 Regressions on other liquidity proxies

The paper studies whether the noise measure is just a repackaging of existing liquidity variables using regressions of the form

$$\Delta\text{Noise}_t = a + b'\Delta X_t + \varepsilon_t, \quad (4)$$

where X_t includes variables such as on-the-run premiums, RefCorp spreads, VIX, default spreads, repo rates, LIBOR spreads, stock returns, and Pastor–Stambaugh liquidity.

Interpretation. If existing variables fully explained changes in Treasury noise, then the measure would add little. The paper finds that these variables are related to noise but leave a large share of its variation unexplained.

3.7 Hedge fund noise beta

For each hedge fund i , the paper estimates exposure to monthly changes in noise using

$$R_t^i = \beta_0 + \beta_i^N \Delta\text{Noise}_t + \beta_i^M R_t^M + \varepsilon_t^i, \quad (5)$$

where:

- R_t^i is the excess return of hedge fund i ;

- ΔNoise_t is the monthly change in the Treasury noise measure;
- R_t^M is the excess return on the CRSP value-weighted stock market portfolio;
- β_i^N is the fund's liquidity-risk exposure.

Interpretation. Since noise rises in bad liquidity states, a negative β_i^N means the fund loses money when liquidity deteriorates. Such funds are more exposed to aggregate liquidity risk.

3.8 Noise-beta-sorted portfolios

Each month, hedge funds are sorted into 10 portfolios using their pre-ranking noise betas. The post-ranking portfolio beta is estimated from

$$R_t^p = \beta_0 + \beta_p^N \Delta\text{Noise}_t + \beta_p^M R_t^M + \varepsilon_t^p, \quad p = 1, \dots, 10. \quad (6)$$

Because hedge fund returns are autocorrelated, the paper also estimates lag-adjusted exposures:

$$R_t^p = \beta_0 + \beta_p^N \Delta\text{Noise}_t + \tilde{\beta}_p^N \Delta\text{Noise}_{t-1} + \beta_p^M R_t^M + \tilde{\beta}_p^M R_{t-1}^M + \varepsilon_t^p. \quad (7)$$

Interpretation. Lagged betas are important because hedge fund returns are often smoothed or serially correlated. The relevant exposure is therefore $\beta_p^N + \tilde{\beta}_p^N$.

3.9 Fama–MacBeth pricing regression

The cross-sectional pricing test is

$$R_t^i = \gamma_{0t} + \gamma_t^N \beta_i^N + \gamma_t^M \beta_i^M + c_t^{age} age_{it} + c_t^{AUM} AUM_{it} + \varepsilon_t^i. \quad (8)$$

The risk premiums are the time-series averages of γ_t^N and γ_t^M .

Interpretation. This asks whether hedge funds with higher exposure to liquidity shocks earn higher average returns, controlling for market exposure, fund age, and fund size.

3.10 Currency carry trade return

For a U.S. investor, the log excess return from holding a foreign currency through a forward contract is

$$rx_{t+1} = f_t - s_{t+1} = i_t^* - i_t + s_t - s_{t+1} = i_t^* - i_t - \Delta s_{t+1}, \quad (9)$$

where:

- f_t is the log forward exchange rate;
- s_t is the log spot exchange rate;
- $i_t^* - i_t$ is the foreign-minus-U.S. interest-rate differential.

Interpretation. High-interest-rate currencies earn carry returns but can crash in bad liquidity states. The paper tests whether their returns are compensation for liquidity risk.

3.11 Carry portfolio risk exposure

For carry portfolio i , the paper estimates

$$R_t^i = \beta_0 + \beta_i^N \Delta \text{Noise}_t + \beta_i^M R_t^M + \varepsilon_t^i. \quad (10)$$

Interpretation. If target currencies have negative noise betas, they lose value when market-wide liquidity worsens. Then part of their high average return can be interpreted as compensation for liquidity risk.

4 Identification and instruments (what makes the estimates credible)

4.1 Why Treasury noise can proxy for aggregate liquidity

- The U.S. Treasury market is highly liquid and default-free, so large deviations from the fitted curve are unlikely to be ordinary credit-risk effects.
- Treasury securities are central collateral in financial markets, so stress in this market is closely connected to aggregate funding and liquidity conditions.
- Individual Treasury-specific demand shocks should often average out across many bonds.
- A market-wide shortage of arbitrage capital, by contrast, should affect the whole curve and increase common dispersion across bonds.

Key identifying logic. The measure is informative when noise reflects broad limits to arbitrage rather than isolated quotation errors or local demand shocks.

4.2 Why the measure is not just poor curve fitting

- The curve-fitting model is flexible enough to capture standard level, slope, and curvature movements.

- The paper checks that noise is not mechanically explained by Treasury rate levels, term spreads, or bond-market volatility.
- The Appendix reconstructs noise using cubic spline methods and finds similar results.
- Figure 1 visually shows that crisis-day noise comes from broad dispersion of market-observed bond yields around the fitted curve, not from a visibly unreasonable curve.

4.3 Why other liquidity variables are useful comparisons

- On-the-run premiums are natural Treasury-market liquidity proxies, but they focus on only a few points on the curve.
- RefCorp spreads capture Treasury flight-to-liquidity premiums.
- VIX and default spreads are broad crisis indicators.
- Pastor–Stambaugh liquidity captures equity-market liquidity.
- Repo and LIBOR spreads capture funding-market stress.

Interpretation. If the Treasury noise measure remains useful after comparison with these variables, then it contains incremental information about market-wide liquidity.

4.4 Why hedge funds are natural test assets

- Hedge funds deploy arbitrage capital across many asset classes.
- Their strategies are often directly exposed to funding and liquidity conditions.
- They are not confined to one market, making them better test assets for market-wide liquidity than ordinary stock or bond portfolios.
- Sorting hedge funds by noise beta creates portfolios with clear differences in liquidity-risk exposure.

4.5 Why currency carry is a natural external validation

- Carry trades are known to earn high average returns but suffer in crash and funding-stress episodes.
- High-interest-rate currencies are often held by arbitrageurs and global investors using leverage.
- If target currencies have high exposure to Treasury noise shocks, this supports the interpretation that noise captures a broad liquidity factor rather than a Treasury-only phenomenon.

4.6 Limitations and cautions

- The noise measure is a proxy, not a direct observation of arbitrage capital.
- The Fama–MacBeth tests use hedge fund data, which may contain reporting, smoothing, and survivorship issues. The paper addresses these, but they remain natural concerns.
- Treasury-market microstructure can change over time, especially around the 1996 change in CRSP quote sources and during 2008–2009.
- The risk premium estimates are economically meaningful but rely on factor-model assumptions about hedge fund returns.
- Currency carry tests use the hedge-fund-estimated noise premium rather than estimating the premium only from the six carry portfolios, which is reasonable but should be interpreted with caution.

4.7 Relation to the literature

- **Limits of arbitrage.** The paper gives empirical content to the idea that scarce arbitrage capital permits price deviations from fundamentals.
- **Liquidity risk pricing.** It extends liquidity asset-pricing work by constructing a market-wide liquidity factor from the Treasury market and testing it outside Treasuries.
- **Hedge fund returns.** It shows that hedge fund performance partly reflects compensation for exposure to market-wide liquidity risk.
- **Currency carry.** It connects the profitability of carry trades to exposure to liquidity crises rather than only to traditional exchange-rate risk.

5 Tables and figures

5.1 Table I: CRSP Treasury data summary statistics

Read:

- Table I describes the raw material behind the noise index.
- On average, there are **163** securities for curve fitting and **109** bonds for noise construction.
- The average bid-ask yield spread is **2.11 basis points**, close to the normal-time average noise level of **3.61 basis points**.
- The average maturity and duration of the bonds used for noise construction are moderate, about **3.85** and **3.29** years.

- The table also shows time variation in available bonds: the late-1990s and early-2000s dip in the number of Treasury securities lines up with reduced Treasury issuance during federal budget surpluses.
- The key message is that the noise measure is constructed from a broad cross-section of ordinary Treasury securities, not from a small number of special bonds.

Table I
CRSP Treasury Data Summary Statistics

Bonds with maturity ranging from (1 month to 10 years) are used for yield curve fitting, while bonds with maturity ranging from (1 year to 10 years) are used to construct the noise measure. All other variables are reported for the sample of bonds used to construct the noise measure, and reported are the time-series averages of the cross-sectional mean, median, and standard deviation. The size of a bond is its amount outstanding in billions of dollars. The bid ask spread is the bid yield minus the ask yield.

Sample Period	# Bonds (1M-10Y)	# Bonds (1Y-10Y)	Coupon (%)	Size (\$B)	Bid/Ask (bps)	Maturity (year)	Age (year)	Duration (year)	Price (\$)	Yield (%)
Mean										
1987-1990	170	115	9.37	7.64	3.99	3.89	3.43	3.15	103.27	8.06
1991-1995	185	126	7.88	10.35	2.61	3.88	3.68	3.21	106.48	5.85
1996-2000	171	111	7.16	12.72	1.93	3.63	4.57	3.09	104.71	5.74
2001-2005	111	66	5.19	19.83	1.27	3.65	4.12	3.24	105.03	3.23
2006-2011	178	124	4.05	24.22	1.30	4.14	3.89	3.68	105.87	2.67
All	163	109	6.52	15.61	2.11	3.85	3.96	3.29	105.18	4.90
Median										
1987-1990			8.94	7.51	3.59	3.42	2.41	2.95	101.75	8.07
1991-1995			7.68	9.84	2.09	3.32	2.47	2.96	104.36	5.87
1996-2000			6.37	12.98	1.63	3.02	2.81	2.72	101.41	5.74
2001-2005			4.84	20.05	1.02	2.94	2.44	2.72	103.60	3.13
2006-2011			3.81	23.54	1.03	3.55	1.87	3.33	103.30	2.57
All			6.12	15.42	1.77	3.25	2.38	2.95	102.95	4.86
Standard deviation										
1987-1990			2.10	3.76	1.96	2.26	3.26	1.53	6.23	0.24
1991-1995			2.05	5.66	1.53	2.36	3.68	1.63	9.32	0.55
1996-2000			2.31	7.76	1.06	2.25	4.84	1.66	8.93	0.15
2001-2005			2.20	9.21	0.75	2.42	4.71	1.94	6.16	0.60
2006-2011			1.87	10.52	0.87	2.50	5.86	1.98	10.47	0.55
All			2.10	7.65	1.19	2.37	4.57	1.77	8.40	0.43

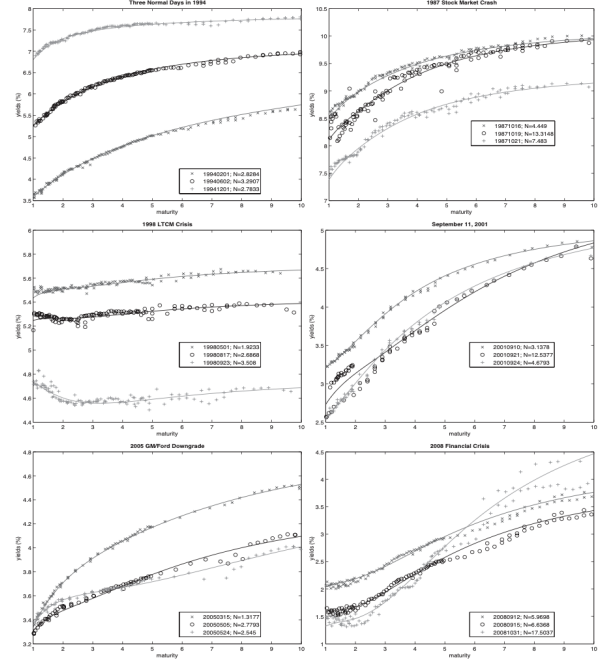


Figure 1. Examples of par-coupon yield curves and the market-observed bond yields, marked by “x”, “o”, or “+”. The top left panel plots three random days in 1994, and the other five panels focus on the days surrounding five events: the 1987 stock market crash, the 1998 LTCM crisis, the September 11, 2001 terrorist attack, the 2005 GE/Ford downgrade, and the Lehman default in September 2008. Marked in the legends are the date of observation and the level of the noise measure for that day.

5.2 Figure 1: examples of fitted curves and observed bond yields

Read:

- Figure 1 plots fitted par-coupon yield curves and market-observed bond yields.
- Normal days in 1994 show low dispersion around the curve.
- Crisis episodes show much larger dispersion: the 1987 crash, LTCM, September 11, the GM/Ford downgrade, and the 2008 crisis.
- This is important because the figure visually supports the interpretation that high noise is not just a curve-fitting artifact.
- The crisis-day yield observations are spread out around the curve across maturities, which is what the paper interprets as evidence of weaker arbitrage activity across the yield curve.

5.3 Figure 2: daily noise over 1987–2011

Read:

- Figure 2 is the paper's main descriptive picture.
- The average noise level is about **3.61 basis points**, with a standard deviation of about **2.17 basis points**.
- Noise is very persistent: daily autocorrelation is about **98.11 percent**, and monthly autocorrelation is about **90.75 percent**.
- During normal periods, noise is comparable to the average bid-ask spread, suggesting that arbitrage normally keeps deviations within transaction-cost bounds.
- During crises, noise jumps sharply: over **13 bps** around the 1987 crash, **5.89 bps** around LTCM, **12.54 bps** after September 11, **8.08 bps** around Bear Stearns, and above **15 bps** after Lehman.
- The main message is that Treasury noise tracks market-wide liquidity crises of very different origins.

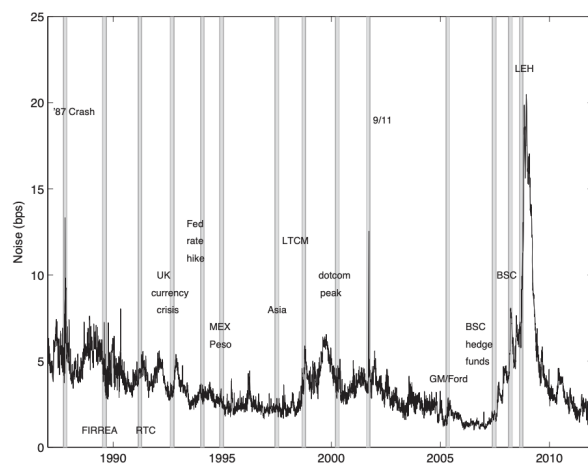


Figure 2. Daily time-series of the noise measure (in basis points). FIRREA: the Financial Institutions Reform, Recovery, and Enforcement Act of 1989; RTC: the Resolution Trust Corporation.

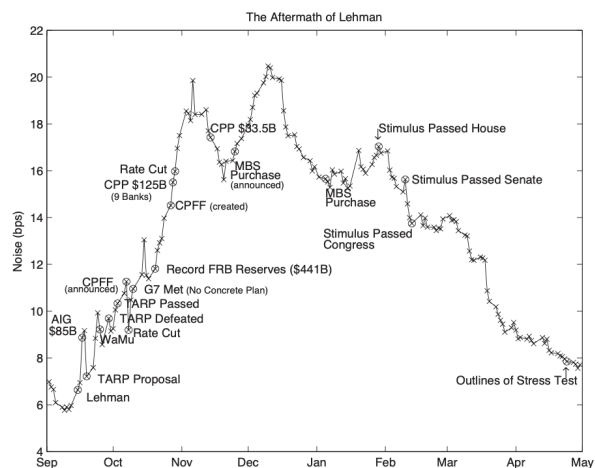


Figure 3. Daily time-series of the noise measure in late 2008 and early 2009. TARP: Troubled Asset Relief Program; CPP: Capital Purchase Program; CPFF: Commercial Paper Funding Facility; and the MBS Program is the Fed's \$1.25 trillion program to purchase agency mortgage-backed securities.

5.4 Figure 3: the aftermath of Lehman

Read:

- Figure 3 zooms in on late 2008 and early 2009.
- On the Lehman default date, noise rises only mildly relative to the severity of the event, from about **5.97** to **6.64** basis points.

- The important point is that the crisis then unfolds gradually: noise stays elevated and peaks much later.
- The first major peak is about **19.85 bps** on November 6, 2008.
- The second peak is about **20.47 bps** on December 10, 2008.
- Noise declines meaningfully only as large policy responses and stress-test announcements help stabilize the financial system.
- The figure shows that the noise measure captures the dynamics of liquidity stress, not merely the announcement-day shock.

5.5 Table II and Table III: relation to other market variables

Table II
Monthly Changes in Noise Measure Regressed on Other Market Variables

Reported are OLS regression coefficients with Newey–West t -statistics in squared brackets. On5Y and On10Y are the on-the-run premiums for 5-year and 10-year bonds. TB3M is the 3-month Treasury bill rate. Repo is the overnight general collateral repo rate. LIBOR is the spread of 3-month LIBOR over a 3-month Treasury bill. Default is the yield spread between Baa and Aaa bond indices. VIX is the volatility index from CBOE. RefCorp is the average spread between Treasury and Refcorp zero-coupon bonds. Δ PSLiq is the innovation in the liquidity factor by Pastor and Stambaugh (2003). StockRet is the monthly return on the CRSP value-weighted index. BondV is the annualized return volatility of monthly bond returns calculated from 5-year Treasury yields using a rolling window of 21 business days. Term is the spread of 10- over 1-year Treasury yields.

Treasury: Level, Slope, and Volatility				On-the-Run Premiums and RefCorp			
(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Δ TB3M	-0.823 [-2.64]		-0.439 [-1.67]	Δ On5Y	0.104 [5.55]		0.062 [2.44]
Δ Term		0.010 [2.40]	0.007 [1.63]	Δ On10Y		0.088 [2.49]	0.090 [2.41]
Δ BondV			0.082 [1.57]	Δ RefCorp		0.029 [3.25]	0.028 [3.60]
Adj R ² (%)	4.66	5.64	1.78	Adj R ² (%)	10.83	18.02	6.11
# month	299	299	299	# month	299	299	248

Stock Market: Ret, VIX, and Liquidity				Repo, LIBOR, and Default			
(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
StockRet	-0.070 [-2.82]		-0.040 [-1.98]	Δ Repo	-0.445 [-3.01]		-0.309 [-2.72]
Δ VIX		0.065 [2.96]	0.042 [2.32]	Δ LIBOR		0.007 [3.24]	0.004 [1.37]
Δ PSLiq			-3.10 [-3.35]	Δ Default		0.027 [2.12]	0.029 [2.11]
Adj R ² (%)	11.79	12.13	4.21	Adj R ² (%)	3.43	3.25	13.00
# month	299	297	287	# month	247	299	299

Table III
Pairwise Correlations (in %)

Pairwise correlations are computed using monthly changes from 1987 through 2011 and reported in percentage. See Table II for definitions of variables.

		2	3	4	5	6	7	8	9	10	11	12	13
1	Δ Noise	-22	33	43	15	24	-20	25	19	36	35	-21	-35
2	Δ TB3M		-18	-14	-25	-50	39	-12	-38	-14	-25	27	17
3	Δ On5Y			13	30	17	-15	5	14	-7	29	-23	-25
4	Δ On10Y				-6	14	0	1	6	24	20	-11	-14
5	Δ BondV					21	-24	21	24	-5	29	-30	-12
6	Δ Term						-32	6	12	-6	4	-15	-4
7	Δ Repo							-19	-19	-9	-2	11	-0
8	Δ RefCorp								17	21	5	-23	-7
9	Δ LIBOR									8	25	-17	-22
10	Δ Default										23	-2	-31
11	Δ VIX											-29	-69
12	Δ PSLiq												31
13	StockRet												

Read:

- Table II regresses monthly changes in noise on changes in other market variables.
- Treasury term-structure variables have limited explanatory power: 3-month bill rate changes explain **4.66 percent**, term spread changes **5.64 percent**, and bond volatility changes only **1.78 percent** of monthly noise variation.
- On-the-run premiums are related to noise, but they do not subsume it. The 5-year premium explains **10.83 percent**; the 10-year premium explains **18.02 percent**; RefCorp explains **6.11 percent**; together these explain **32.93 percent**.
- Equity and crisis variables are also related: stock returns, VIX, and Pastor–Stambaugh liquidity jointly explain **16.65 percent**.

- Funding and credit variables are related as expected: repo, LIBOR, and default spreads jointly explain **21.06 percent**.
- When many standard liquidity proxies are considered jointly, they explain about **43.7 percent** of monthly noise variation, leaving more than half unexplained.
- Table III shows the pairwise correlations: changes in noise are positively correlated with on-the-run premiums, RefCorp, default spreads, and VIX, and negatively correlated with stock returns and repo rates.
- The key message is that noise is connected to known liquidity variables but also contains distinct information.

5.6 Table IV: TASS hedge fund sample

Read:

- Table IV describes the hedge fund data used in the pricing tests.
- The full sample contains **5,392** funds, of which **3,557** are graveyard funds.
- The average monthly fund return is **0.67 percent**, and average monthly volatility is **3.68 percent**.
- Average AUM is about **\$160 million**, while median AUM is about **\$56 million**, indicating a right-skewed size distribution.
- Average return autocorrelation is about **0.19**, which motivates the lagged beta specifications later in the paper.
- The table also shows that hedge funds differ substantially by style, which is why fund-level tests are preferable to style-index tests.

	Total		Graveyard		ret (%)		stdret(%)		AUM(EM)		iAUM(EM)		Reporting (mn)		Age (mn)		Auto Corr	
	(#)	(#)	Mean	Med	Mean	Med	Mean	Med	Mean	Med	Mean	Med	Mean	Med	Mean	Med	Mean	Med
Panel A: All Hedge Funds																		
1994-1999	1,856	1,433	1.81	1.23	4.47	3.52	78.10	26.74	16.58	5.40	129.51	133.00	29.91	21.50	0.11	0.13		
2000-2006	4,602	3,201	0.87	0.76	2.93	2.07	136.02	49.44	22.32	8.18	92.78	79.00	42.59	28.00	0.12	0.13		
2007-2011	4,081	2,246	0.21	0.21	3.83	2.94	234.64	68.22	29.49	10.00	88.73	74.00	75.11	59.50	0.17	0.18		
All	5,392	3,557	0.67	0.59	3.68	2.78	160.21	55.70	36.89	9.71	84.65	70.00	46.34	37.50	0.19	0.19		
Panel B: Hedge Funds by Style																		
Long/short equity	1,393	1,005	0.88	0.80	4.73	3.98	116.19	49.09	16.92	6.47	86.30	72.00	46.97	38.00	0.13	0.13		
Global macro	215	145	0.73	0.65	4.10	3.35	318.71	53.85	42.20	8.67	75.07	63.00	41.87	33.00	0.07	0.08		
Fund of funds	1504	986	0.39	0.38	2.58	2.06	156.08	55.26	36.34	12.29	87.75	75.00	48.19	40.00	0.24	0.25		
Fixed income Arb	170	134	0.58	0.59	2.39	2.03	195.14	93.95	27.83	10.67	80.56	71.00	42.21	37.00	0.22	0.19		
Managed futures	275	140	0.79	0.72	5.10	4.41	178.14	45.65	21.43	5.03	101.04	76.00	57.83	43.00	0.03	0.03		
Event driven	459	339	0.82	0.73	2.76	2.30	226.59	89.16	25.46	7.80	92.68	80.00	50.49	41.50	0.25	0.24		
Equity neutral	242	191	0.54	0.45	2.61	2.15	98.02	43.98	25.48	9.34	72.31	60.00	38.04	31.25	0.11	0.12		
Emerging markets	436	294	0.80	0.77	5.83	5.20	121.15	48.85	27.71	10.70	76.12	63.00	42.20	34.50	0.21	0.22		
Convertible Arb	153	123	0.57	0.58	2.66	1.99	171.95	68.65	19.92	10.00	91.18	82.00	49.04	41.00	0.38	0.43		
Others	545	320	0.69	0.62	3.33	2.69	201.33	64.42	32.11	10.16	72.41	55.00	39.66	30.00	0.23	0.22		

Rank	exret (%)	Pre Formation				Post Formation				Alpha (%)			
		Δ Noise ρ^2	Mkt ρ^2	Adj R ² (%)	Δ Noise ρ^2	Mkt ρ^2	Adj R ² (%)	$\rho^2 + \text{lag}$	Mkt $\rho^2 + \text{lag}$	Adj R ² (%)	No lag ρ	With lag ρ	
1	0.95 [3.51]	1.20 [4.39]	-2.96 [-42.68]	0.55 [33.62]	-0.64 [-5.44]	0.49 [10.67]	52.77 [2.53]	-1.90 [-5.53]	0.55 [9.71]	55.19 [0.05]	0.01 [0.05]	0.05 [0.20]	
2	0.60 [3.50]	0.85 [4.88]	-1.20 [-41.74]	0.37 [55.01]	-0.41 [-3.81]	0.33 [11.14]	58.76 [4.10]	-0.55 [-4.10]	0.39 [10.22]	61.28 [0.08]	-0.01 [-0.08]	0.03 [0.15]	
3	0.48 [3.49]	0.72 [5.24]	-0.72 [-36.86]	0.28 [31.47]	-0.25 [-2.53]	0.26 [10.53]	54.45 [2.09]	-0.31 [-2.09]	0.32 [10.99]	57.79 [0.29]	0.04 [0.36]	0.05 [0.36]	
4	0.43 [3.39]	0.67 [5.26]	-0.47 [-28.51]	0.25 [32.13]	-0.28 [-3.04]	0.23 [9.81]	54.46 [2.39]	-0.36 [-2.39]	0.29 [9.24]	58.50 [0.05]	-0.01 [-0.05]	0.01 [0.07]	
5	0.34 [3.01]	0.58 [5.12]	-0.29 [-19.16]	0.22 [21.25]	-0.30 [-3.21]	0.20 [9.20]	52.62 [2.43]	-0.36 [-2.43]	0.25 [8.40]	56.83 [0.63]	-0.07 [-0.63]	-0.04 [-0.37]	
6	0.38 [3.76]	0.63 [6.02]	-0.14 [-9.14]	0.21 [35.25]	-0.25 [-3.16]	0.19 [9.48]	56.95 [2.61]	-0.34 [-2.61]	0.24 [8.98]	61.70 [0.14]	0.01 [0.14]	0.03 [0.27]	
7	0.29 [2.87]	0.53 [5.21]	0.02 [1.01]	0.21 [35.28]	-0.24 [-3.17]	0.19 [9.33]	56.44 [1.69]	-0.19 [-1.69]	0.24 [8.38]	59.39 [0.70]	-0.07 [-0.70]	-0.02 [-0.15]	
8	0.38 [3.51]	0.62 [5.68]	0.24 [10.02]	0.24 [36.02]	-0.20 [-2.17]	0.21 [9.76]	55.24 [0.35]	-0.04 [-0.35]	0.26 [9.54]	57.49 [0.30]	0.03 [0.30]	0.10 [0.94]	
9	0.36 [3.75]	0.60 [4.59]	0.63 [17.09]	0.29 [33.47]	-0.08 [-0.89]	0.26 [9.31]	55.41 [0.57]	0.08 [0.57]	0.31 [9.32]	56.58 [0.23]	0.03 [0.23]	0.07 [0.53]	
10	0.23 [1.21]	0.47 [2.49]	1.94 [25.49]	0.40 [23.18]	0.15 [0.62]	0.33 [6.76]	35.09 [2.60]	0.15 [2.60]	0.33 [8.08]	39.06 [0.01]	0.00 [-0.01]	0.03 [0.16]	

5.7 Table V, Table VI, and Figure 4: noise-beta-sorted hedge fund portfolios

Read:

- Table V sorts hedge funds into 10 portfolios based on pre-ranking noise beta.
- Portfolio 1 has the most negative noise beta and is most exposed to bad liquidity states.
- Portfolio 10 has the most positive noise beta and is least exposed to liquidity stress.
- Portfolio 1 earns average excess returns of about **0.95 percent per month**; portfolio 10 earns about **0.23 percent per month**.
- The spread is about **0.72 percent per month**, consistent with compensation for liquidity-risk exposure.
- Post-ranking betas confirm the sorting: portfolio 1 has post-ranking noise beta around **-0.64**, while portfolio 10 has about **0.15**; with lagged betas the spread is even larger, from about **-1.00** to **0.69**.
- Table VI shows that the extreme portfolios are not simply explained by obvious fund characteristics such as AUM, age, or reporting length.
- Figure 4 shows that in 2008, the exit rate of the high-liquidity-risk portfolio rises especially sharply, supporting the idea that these funds were genuinely exposed to liquidity crises.

Table VI
Noise-Beta-Sorted Portfolios, Characteristics

The 10 portfolios are ranked by their noise betas. See Table IV for variable definitions.

Portfolio Rank	1	2	3	4	5	6	7	8	9	10
Panel A: Characteristics										
AUM (\$M)	163	183	205	217	218	212	200	187	170	160
iAUM (\$M)	21.08	19.61	22.18	21.64	21.23	20.13	20.80	19.67	19.91	17.41
reporting (mn)	130	128	132	135	134	135	133	133	129	130
age (mn)	75.9	75.6	76.3	78.0	78.0	77.7	76.9	77.0	76.4	77.0
stdret (%)	3.77	2.40	1.90	1.76	1.56	1.44	1.41	1.51	1.80	2.61
auto corr	0.15	0.19	0.22	0.23	0.25	0.25	0.23	0.20	0.16	0.11
Panel B: Allocation within Hedge Fund Style (%)										
Long/short equity	11.99	12.08	9.16	7.22	6.08	6.27	7.51	10.16	13.85	15.67
Global macro	16.65	13.32	9.62	6.38	5.25	4.88	6.85	9.63	13.68	13.74
Fund of funds	4.12	7.62	11.48	14.73	15.75	14.82	12.61	9.46	5.98	3.42
Fixed income Arb	10.34	7.28	9.10	11.29	11.65	11.71	11.40	11.73	9.71	5.80
Managed futures	17.79	9.80	5.41	4.17	3.92	4.13	5.46	8.16	13.76	27.41
Event driven	4.87	8.33	10.91	11.16	12.23	12.91	12.62	12.65	8.68	5.63
Equity neutral	3.69	7.30	9.94	8.15	7.90	9.61	11.95	13.83	16.85	10.77
Emerging markets	29.62	15.91	9.86	6.30	4.87	4.54	5.12	6.22	7.57	10.00
Convertible Arb	11.33	12.71	11.93	11.18	12.42	12.70	12.39	8.48	5.12	1.73
Others	6.76	8.67	10.56	10.41	9.79	11.27	12.77	11.06	9.99	8.72

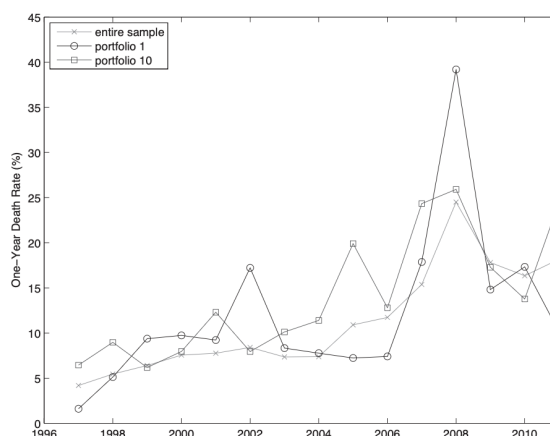


Figure 4. The year t exit rate of hedge funds in the top- and bottom-ranked noise-beta-sorted portfolios.

5.8 Table VII: Fama–MacBeth pricing tests using hedge fund returns

Read:

- Table VII is the main asset-pricing table.
- Using contemporaneous noise beta, the estimated liquidity risk premium is **-0.69 percent per month**, with a t-statistic of **-2.37**.
- Using contemporaneous plus lagged beta, the estimated premium is **-0.35 percent per month**, with a t-statistic of **-2.52**.

- The sign is negative because higher noise means worse liquidity. A fund with a more negative noise beta loses more in bad states and therefore earns higher average returns.
- The result is robust when noise is scaled by bid-ask spreads and when the component of noise explained by VIX and default spreads is removed.
- Other liquidity proxies - on-the-run premiums, RefCorp, Pastor–Stambaugh liquidity, VIX, and default spreads - are not significantly priced in the same hedge fund tests.
- The main message is that Treasury noise is not merely another crisis proxy; it has unique pricing power for hedge fund returns.

Table VII
Estimating Liquidity Risk Premiums Using Hedge Fund Returns

Each proxy of liquidity is tested together with the equity market portfolio in a two-factor model using hedge fund returns, with age and size (AUM) as additional controls. The coefficients for age are reported in percentage points. The Fama–MacBeth *t*-statistics are reported in square brackets. Panel A focuses on the noise measure with the base case as described in equations (5) and (7) as well as additional cases. Panel B considers other proxies for liquidity. See Table II for variable definitions

Factor	Intercept	Liquidity	Market	Age	AUM
Panel A: Noise as Proxy for Liquidity					
Noise	1.73 [4.39]	-0.69 [-2.37]	1.01 [1.65]	0.015 [0.26]	-0.10 [-3.96]
Noise (beta+lag beta)	1.77 [4.37]	-0.35 [-2.52]	1.01 [1.8]	0.016 [0.29]	-0.10 [-3.95]
Noise/BASpreads	1.75 [4.38]	-0.56 [-2.20]	0.89 [1.49]	0.009 [0.16]	-0.10 [-3.93]
Noise-VIX-Default	1.65 [4.14]	-1.19 [-2.26]	0.97 [1.67]	0.008 [0.15]	-0.10 [-3.88]
Panel B: Other Proxies for Liquidity					
On5Y	2.00 [4.46]	-1.58 [-0.79]	0.63 [0.93]	0.003 [0.04]	-0.10 [-3.96]
On10Y	1.80 [4.03]	-0.70 [-0.25]	1.05 [1.76]	0.002 [0.03]	-0.09 [-3.8]
RefCorp	1.77 [4.37]	-5.32 [-1.27]	0.73 [1.22]	0.017 [0.31]	-0.10 [-4.01]
PSLiq	1.87 [4.51]	-0.02 [-0.22]	0.84 [0.96]	-0.032 [-0.59]	-0.09 [-3.89]
VIX	1.79 [4.23]	1.92 [0.49]	0.65 [0.72]	-0.004 [-0.08]	-0.09 [-3.83]
Default	1.88 [4.19]	8.27 [1.37]	1.17 [1.79]	-0.015 [-0.28]	-0.09 [-3.89]

Portfolios are formed monthly by sorting currencies by their forward discount, with currencies in portfolio 1 having the highest forward discount and the highest interest rate and currencies in portfolio 6 having the lowest interest rate. Returns are monthly in excess of the risk-free rate. The two-factor model includes Δ Noise and stock market returns, and the two-factor alphas are calculated using the factor risk premiums estimated using hedge fund returns. Also reported are the CAPM beta and alpha as well as the one-factor noise beta. Robust Newey–West <i>t</i> -statistics are reported in square brackets.								
Rank	exret (%)	Two-Factor Model			CAPM		One Factor	
		Δ Noise β^N	Market β^M	Adj R^2 (%)	Alpha (%)	Beta	Alpha (%)	Δ Noise Beta
Panel A: Developed and Emerging Countries								
1	0.79 [4.56]	-0.57 [-1.84]	0.15 [2.64]	11.10	0.24 [1.38]	0.19 [3.08]	0.69 [3.22]	-0.83 [-2.51]
2	0.35 [2.39]	-0.29 [-1.54]	0.15 [2.90]	10.18	-0.01 [-0.07]	0.17 [3.64]	0.26 [1.55]	-0.54 [-2.80]
3	0.28 [2.14]	-0.32 [-1.41]	0.09 [1.98]	6.96	-0.04 [-0.34]	0.12 [2.36]	0.22 [1.39]	-0.48 [-1.93]
4	0.15 [1.21]	-0.12 [-0.61]	0.07 [1.65]	2.84	-0.01 [-0.08]	0.08 [1.91]	0.11 [0.77]	-0.24 [-1.13]
5	-0.05 [-0.38]	-0.14 [-0.66]	0.05 [1.34]	2.19	-0.21 [-1.72]	0.07 [1.52]	-0.08 [-0.58]	-0.23 [-0.97]
6	-0.18 [-1.37]	-0.05 [-0.33]	0.00 [0.06]	0.06	-0.22 [-1.73]	0.01 [0.24]	-0.18 [-1.30]	-0.06 [-0.38]
Panel B: Developed Countries Only								
1	0.53 [2.81]	-0.78 [-3.45]	0.13 [2.06]	11.07	-0.15 [-0.79]	0.19 [2.87]	0.43 [1.98]	-1.01 [-4.33]
2	0.40 [2.35]	-0.69 [-2.51]	0.15 [2.95]	14.78	-0.25 [-1.50]	0.21 [3.37]	0.29 [1.48]	-0.96 [-3.14]
3	0.25 [1.56]	-0.40 [-1.47]	0.12 [2.30]	8.00	-0.16 [-1.00]	0.15 [2.67]	0.17 [0.95]	-0.61 [-2.08]
4	0.23 [1.47]	-0.18 [-0.59]	0.05 [1.01]	1.48	0.05 [0.29]	0.07 [1.26]	0.19 [1.04]	-0.26 [-0.84]
5	0.03 [0.21]	-0.06 [-0.27]	0.02 [0.40]	0.22	-0.05 [-0.32]	0.03 [0.66]	0.01 [0.08]	-0.09 [-0.40]
6	0.00 [0.01]	0.47 [2.04]	0.02 [0.27]	1.73	0.30 [1.63]	-0.01 [-0.19]	0.01 [0.04]	0.44 [2.15]

5.9 Table VIII: currency carry portfolios

Read:

- Table VIII tests whether the hedge-fund-estimated liquidity premium can help explain currency carry trade returns.
- In the broad developed-plus-emerging sample, portfolio 1 - high-interest-rate target currencies - earns about **0.79 percent per month** before risk adjustment.
- Portfolio 1 has a negative noise beta of about **-0.57**, while portfolio 6 - funding currencies - has a small and insignificant beta of about **-0.05**.
- After adjusting for noise and market risk, portfolio 1's alpha falls to about **0.24 percent per month** and is no longer statistically significant.

- By contrast, a CAPM-only adjustment leaves portfolio 1 with a significant alpha of about **0.69 percent per month**.
- In the developed-countries-only sample, the pattern is even cleaner: portfolio 1 has a noise beta of about **-0.78**, and the two-factor alpha becomes insignificant.
- The table's main message is that carry trade profits are closely connected to exposure to aggregate liquidity risk.

5.10 Appendix Figure A1 and Table A.I: robustness to curve fitting

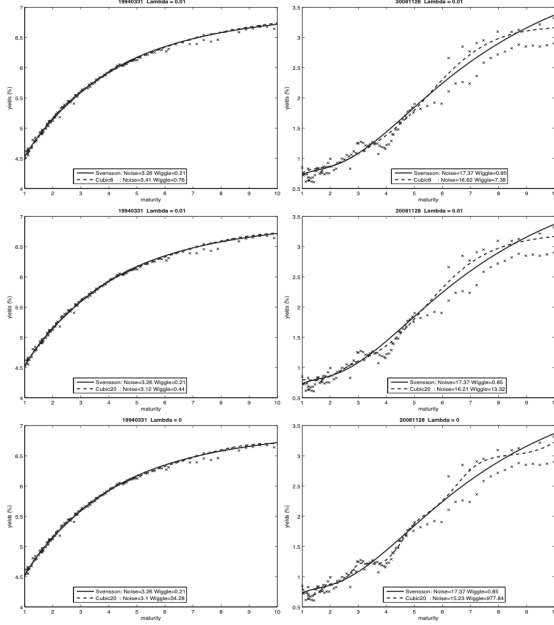


Figure A1. Examples of par-coupon yield curves and the market-observed bond yields, marked by "x." The left panels are for a normal day (March 31, 1994), while the right panels are for a stressful day (November 30, 2008). The Svensson model is the base case in all figures and is marked by solid lines. The cubic-spline models are marked by dashed lines. The first row plots Cubic8 with $\lambda = 0.01$, the second row plots Cubic20 with $\lambda = 0.01$, and the last row plots Cubic20 with $\lambda = 0$. Marked in the legend are the levels of the noise measure and the "wiggles"—the integrated curvature of the forward curve.

Table A.I
Estimating Liquidity Risk Premiums Using Hedge Fund Returns

The noise measure is tested together with the equity market portfolio in a two-factor model using hedge fund returns, with age and size as additional controls. Fama-MacBeth t -statistics are reported in square brackets. The noise measure is constructed using two cubic spline methods: Cubic8 and Cubic20. Reported in the table are the estimated market prices of risk for our noise measure and the equity market portfolio

	Cubic8		Cubic20	
	Liquidity	Market	Liquidity	Market
Panel A: With Fixed Penalty Coefficient $\lambda = 0.01$				
Noise	-0.61 [-2.23]	0.88 [1.51]	-0.49 [-2.14]	1.44 [2.23]
Noise(beta+lag beta)	-0.30 [-2.28]	0.79 [1.57]	-0.29 [-2.26]	1.21 [2.16]
Noise/BASpreads	-0.46 [-1.97]	1.00 [1.74]	-0.40 [-1.99]	1.23 [1.95]
Noise-VIX-Default	-1.19 [-2.12]	1.27 [2.06]	-1.17 [-2.11]	2.50 [2.97]
Panel B: With Varying Penalty Coefficient λ				
$\lambda = 0$	-0.83 [-1.70]	0.76 [1.22]	-0.33 [-1.68]	1.10 [1.83]
$\lambda = 0.005$	-0.71 [-2.17]	0.81 [1.37]	-0.45 [-1.94]	1.54 [2.39]
$\lambda = 0.01$	-0.61 [-2.23]	0.88 [1.51]	-0.49 [-2.14]	1.44 [2.23]
$\lambda = 0.02$	-0.53 [-2.19]	0.74 [1.25]	-0.47 [-2.10]	1.34 [2.04]
$\lambda = 0.03$	-0.50 [-2.08]	0.77 [1.28]	-0.51 [-2.22]	1.19 [1.86]
$\lambda = 0.04$	-0.49 [-2.05]	0.76 [1.23]	-0.48 [-2.10]	1.18 [1.83]
$\lambda = 0.05$	-0.48 [-2.00]	0.79 [1.26]	-0.44 [-2.02]	1.19 [1.81]

Read:

- The Appendix reconstructs the noise measure using cubic spline methods instead of the Svensson model.
- Figure A1 shows that alternative curve-fitting methods still identify large crisis-period dispersion in Treasury yields.
- Table A.I shows that the asset-pricing conclusions are robust under alternative curve-fitting choices.
- This is important because the paper's main measure could otherwise be criticized as an artifact of the chosen yield-curve model.

6 Short summary

- **Method.** The paper fits a smooth Treasury yield curve each day and uses the root mean squared yield deviation from that curve as a market-wide noise measure.
- **Interpretation.** The measure is interpreted as an inverse proxy for aggregate arbitrage capital: when arbitrage capital is scarce, Treasury yields become more misaligned across the curve.
- **Time-series evidence.** Noise spikes in major crises and is related to, but not subsumed by, on-the-run premiums, RefCorp spreads, VIX, default spreads, repo rates, LIBOR spreads, stock returns, and equity liquidity.
- **Asset-pricing evidence.** Hedge funds with more negative noise betas earn higher average returns, and the noise factor is significantly priced in Fama–MacBeth tests.
- **External validation.** Currency carry trade profits are much smaller after adjustment for exposure to the noise factor, especially for high-interest-rate target currencies.

7 Conclusion

7.1 Core conclusion

- The paper shows that cross-sectional pricing errors in U.S. Treasuries contain useful information about aggregate market liquidity.
- Treasury noise captures crises of different origins and magnitudes, suggesting that it reflects broad arbitrage-capital conditions rather than local Treasury-market illiquidity alone.
- The noise factor helps explain hedge fund returns and currency carry returns, while many standard liquidity proxies do not have the same pricing power in these tests.

7.2 Economic intuition

- In normal times, relative-value arbitrageurs keep the Treasury curve smooth.
- In crises, arbitrageurs lose capital, face tighter funding, or become unwilling to deploy capital.
- The same bond-specific liquidity shocks that would normally be arbitrated away then generate visible dispersion in yields.
- Because major arbitrageurs operate across markets, Treasury noise also reveals stress in broader financial markets.
- Assets and strategies that lose when noise rises must offer higher expected returns as compensation.

7.3 Takeaway

This paper is important because it turns a traditional pricing residual into an economically meaningful state variable. The central contribution is not merely that Treasury prices sometimes deviate from a fitted curve. The contribution is the interpretation and validation of those deviations: they summarize the market-wide scarcity of arbitrage capital and help price returns in hedge funds and currency carry trades. In this sense, the paper shows how “noise” can become information.